

Understanding RSSI

RSSI stands for **Received Signal Strength Indicator**. It is a measurement of the power present in a radio signal that a device receives, typically expressed in decibel-milliwatts (**dBm**). A higher RSSI value (closer to 0) indicates a stronger signal, while negative values are common, with numbers like -55 dBm being stronger than -75 dBm.

Signal Strength:

- RSSI provides a general indication of how strong the received radio signal is from nearby tags.

Negative Values:

- RSSI values are typically negative, as they represent a loss of power relative to a milliwatt.

Interference:

- RSSI can be influenced by noise and interference from other sources on the same frequency, not just the desired signal.
- RSSI values can vary among tags within the same category due to differences in individual tag performance.

RSSI (dBm)	Signal Strength	What It Means
-11 to -30	Very Strong	Tag is very close
-31 to -50	Strong	Tag is close
-51 to -70	Medium	Tag is nearby
-71 to -90	Weak	Tag is far or partially blocked

BB15E0450000007E0CF000000000000000000000000**F841**7BAE

16-bit unsigned integer:

A 16-bit unsigned integer uses 16 binary digits (bits). Each bit can be either 0 or 1, providing 2 possible states per bit. This results in a total of:

$$2^{16} = 65,536$$

Since counting starts from 0, the maximum representable value is:

$$2^{16} - 1 = 65,535$$

Thus, a 16-bit unsigned integer can represent values in the range **0 to 65,535**.

Manual calculation of RSSI:

Hexadecimal value = F841

16-bit signed int (Decimal) = 16-bit unsigned int. - Max 16-bit unsigned int.

$$= 65535 - 65536 = \mathbf{-1983}$$

RSSI calculation using a Calculator:

In the Programmer Calculator, select the "**Word**" data size, then enter the RSSI hexadecimal value to convert it to its decimal equivalent.



